

TPC-416: Four-Shell Odd Pooled Extension

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September 6, 2026

Abstract

We add the complete shell at $Q = 524288$ to the pooled profile. The four shells retain 75483 primes. At $H = 66$, $N = 264$, shell-local amplitudes and pooled alternating CRT give the explicit odd parity counts $m_- = 37741, m_+ = 37742$. Exact arithmetic and an independent literal replay verify the finite row; no arithmetic theorem is claimed.

1 Four-shell profile

Let $\mathcal{P}$ be the increasing union of four consecutive dyadic shells from 65536 through 1048576. Set $a_i = p_i^3/[Q_i^2(p_i - 1)]$ using each source-shell scale, and impose alternating CRT residues with $H = 66, N = 264 = 4H$. The total count is $5709 + 10749 + 20390 + 38635 = 75483$, so $m_- = 37741, m_+ = 37742$. Every prime exceeds N , hence

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

The same Cauchy–Schwarz argument gives

$$0 \leq z \leq \frac{t_1}{a_{\min} \sqrt{S_0 S_1}} \leq \frac{4}{a_{\min} H} \leq \frac{4}{H}. \quad (1)$$

2 Exact observation

The certificate stores all 75483 prime labels, shell scales, CRT data, and exact rational energies. The independent checker rebuilds the four sieves and literally visits every mask for every prime and coordinate before comparing G_0 , G_1 , and M .

3 Scope and reproduction

This one finite synthetic adjacent normalized proxy entry does not prove a full operator estimate, physical h_0 or arithmetic signs, pay arithmetic L^2 or fixed-power credit, close Route A/B, or imply a twin-prime result. Run the producer, replay, and mutation stress with `--check` in normal and optimized modes; Bridge-B repeats and locks these checks.